

# Element I

## Testing, data collection and analysis

Through the conduct of several tests for high priority requirements that are reasonable based on instructional contexts, or through physical or mathematical modeling, the student demonstrates considerable understanding of testing procedure, including the gathering and analysis of resultant data; the analysis of the effectiveness with which the design met stated goals includes a consistently detailed explanation [and summary] of the data from each portion of the testing procedure and from expert reviews, generously supported by pictures, graphs, charts and other visuals; the analysis includes an overall summary of the implications of all data for proceeding with the design and solving the problem.

To ensure a proper, in-depth analysis of our final product, the following is our group's testing procedure.

### 1. Preparation Phase:

- Gather all necessary materials and equipment for the testing procedure.
- Ensure the detachable lid and barrel system is securely assembled and ready for testing.
- Set up the testing environment in a safe and controlled area.

### 2. Baseline Measurements:

- Measure the initial weight and dimensions of the barrel system.
- Record the time it takes to securely attach the lid to the barrel.

### 3. Functional Testing:

- Place food items or treats inside the barrel to simulate the feeding process.

- Securely attach the lid to the barrel.
- Observe and record the time it takes for the bears (or bear-mimicking mechanism) to successfully open the lid and access the food.
- Repeat this process multiple times to ensure consistency and reliability of the lid and barrel system.

#### 4. Durability Testing:

- Subject the lid and barrel system to simulated bear-like forces using controlled mechanical pressure or weight.
- Gradually increase the pressure or weight to assess the durability and strength of the system.
- Record any signs of damage, deformation, or failure of the components.

#### 5. Ease of Access Testing:

- Invite individuals of varying ages and strengths (including the high schoolers working on the project) to attempt opening and closing the lid.
- Record any difficulties, challenges, or ease of use encountered during the process.
- Collect feedback on the ergonomics and usability of the detachable lid and barrel system.

#### 6. Data Analysis:

- Compile all recorded measurements, observations, and feedback.
- Analyze the effectiveness of the design based on factors such as:
  - Time taken to open/close the lid.
  - Durability under simulated bear-like forces.
  - Ease of access for users.

The results were the following:

Baseline Measurements (Dimension):

Height: 26.75 inches

Diameter: 19.75 inches

Radius: 9.875 inches

Approximate Container Volume  $\approx$  8194.97943 cubic inches or 35 gallons

Baseline Measurements (Time to Access (TTA)):

Initial Conditions: The person who consistently opened the container was also someone properly trained to efficiently open the container. The container also contained no objects prior to being accessed. There was simulated time added to the total time to simulate a lock secured to the container to be opened the first try. Additionally, the testing stage was rushed due to a small time constraint, limiting the number of trials to two rather than more.

| Different Accessing Occasions in Seconds | Opening a closed container (with lock) | Closing an open container (no lock) |
|------------------------------------------|----------------------------------------|-------------------------------------|
| Trial 1                                  | 8.76                                   | 6.77                                |
| Trial 2                                  | 8.51                                   | 5.48                                |

Bear Simulation (Human Testing):

Two participants, one who was part of the design team (Participant 1) and the other who was not affiliated with the class in any sort of way (Participant 2), agreed to partake in the experiment. Each participant was given the assembled puzzle and were required to take out a certain random number of sliced potatoes out of the total number of potato slices (which were eleven in total) while being timed. This ensured that there will be a uncontrollable factor of the bear that could distract or prevent the bear from continue eating. Each time, the participates were told to think and act like a bear and to roughly handle the puzzle as bear can only roughly grip objects because they have no opposable thumbs. This ensured the realism that would be consistent with actually handling this puzzle off to a bear. Additionally, the participants were told not to directly reach into the puzzle as the bear's arms are significantly larger than the holes that were cut to allow food to pass through the puzzle.

Participant 1 was assigned to empty the puzzle while Participant 2 was assigned 5 pieces to access out of the puzzle. Participant 1 finished the puzzle within 2 minutes and 15 seconds. They reported that they had no trouble tossing the puzzle around and that they did not seem to feel any sharp edges. They also report that they had difficulty have the holes aligned with each other while the puzzle was in movement, but they were later explained to that the aspect of which was intentional to prevent the puzzle from being instantly solved by the bear and to keep the bear occupied.

Participant 2 finished within 1 min and 45 seconds. They also reported that they had no trouble tossing the puzzle and found no sharp or dangerous edges that could possibly harm them. Similarly to Participant 1, Participant 2 also reports the same difficulty aligning the holes. They were later told that it was intentional.

### Bear Simulation Testing (PSI Handling):

During our first meeting with the zookeepers, the zookeeper answered some initial questions, one of which included the bite force of the bear. The zookeepers then simplified our question and provided that any item able to handle the bite force of the bear, which was 600 pounds per square inch (PSI), would be able to withstand the bear itself. Thus, we concluded that if 200 pounds sustained at a quarter of a square inch could then withstand the same amount of force as the bite force of the bear.

The experiment was simple: add aluminum panels until a desired weight is met. We measured the weight of 8 panels, which totaled to 20 pounds and stacked 8 panel increments onto a quarter-square inch on the top most part of the barrel. The barrel handled a total of about 30 to 40 total panels, which equaled around 60 pounds, before running out of panels to stack. Thus, we ask someone who weighed around 150 pounds to sit on top of both the panels for some duration of time.

### Analysis of Testing Procedure and Results:

### Baseline Measurements:

The initial measurements of the barrel system provide essential data for assessing its size and capacity. With a height of 26.75 inches and a diameter of 19.75 inches, the barrel has a volume of approximately 35 gallons. These dimensions serve as a reference point for evaluating the barrel's suitability for containing food items and fitting within the designated habitat space.

### Baseline Measurements (Time to Access):

The time taken to securely attach the lid to the barrel is crucial for evaluating the efficiency of the system. Despite limited trials due to time constraints, the recorded times for opening and closing the container indicate relatively quick access, with both trials averaging under 10 seconds. However, the initial conditions, including the simulated lock and rushed testing stage, may have influenced the results and warrant further investigation.

### Bear Simulation (Human Testing):

The human testing phase aimed to mimic bear behavior and assess the practicality of the puzzle feeder design. Participants, acting as bears, successfully accessed food items from the puzzle within a reasonable timeframe. Both participants reported no difficulties in handling the puzzle and noted the intentional design features, such as misaligned holes, to prolong engagement. This positive feedback suggests that the puzzle feeder effectively engages bears while ensuring safety and usability.

### Bear Simulation Testing (PSI Handling):

The PSI handling test evaluates the durability of the barrel system against bear-like forces. By simulating the bear's bite force using weighted aluminum panels, the barrel demonstrated exceptional strength, withstanding up to 150 pounds of human weight plus additional weights. This exceeds the anticipated force requirements, indicating the barrel's robustness against vertical pressure.

### Overall Implications:

The testing procedure provides comprehensive insights into the effectiveness of the detachable lid and barrel system. Despite some limitations, such as the small number of trials in certain tests, the results indicate promising performance across key criteria. The barrel's size and capacity are suitable for containing food items, while the quick access time and durability ensure practicality and safety for bear interaction.

Additionally, positive feedback from human participants underscores the design's success in engaging bears while considering their natural behavior and physical capabilities. The intentional design features, such as misaligned holes, contribute to prolonging engagement and enrichment.

Furthermore, the barrel's exceptional durability against bear-like forces confirms its suitability for withstanding the rigors of habitat use. The PSI handling test demonstrates that the barrel exceeds anticipated force requirements, ensuring resilience and longevity in real-world scenarios.

In conclusion, based on the analysis of testing data, the detachable lid and barrel system appears to meet stated goals effectively, offering a viable solution for enriching the habitat of Andean bears. Further refinement and testing may be necessary to address any identified limitations and optimize the design for practical implementation.